

step 3: replace terminals in long productions

CNF requires:

- Either $A \rightarrow BC$
- or $A \rightarrow a$

$$X_a \rightarrow a \quad X_b \rightarrow b$$

update rules:

$$B \rightarrow cb \Rightarrow B \rightarrow CX_b$$

$$D \rightarrow bD \Rightarrow D \rightarrow X_b D$$

$$A \rightarrow bD \Rightarrow A \rightarrow X_b D$$

step 4: Convert long productions to binary form

ex: $S \rightarrow ABCBCDA$

let

$$S \rightarrow AS_1$$

$$S_1 \rightarrow BS_2$$

$$S_2 \rightarrow CS_3$$

$$S_3 \rightarrow BS_4$$

$$S_4 \rightarrow CS_5$$

$$S_5 \rightarrow DA$$

final $\Rightarrow$

$$A \rightarrow CD \mid a \mid X_b D \mid b$$

$$B \rightarrow CX_b \mid b$$

$$C \rightarrow a$$

$$D \rightarrow X_b D \mid b$$

Ques 15) Apply the rules and show step by step conversion of the following grammar to CNF.